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Auto LIBRIS- A comprehensive digital library resource management framework using structures, pointers, and file operations

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SUBJECT:- C- Programming

CODE:- CSA0201

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ABSTRACT:

- AutoLibris is an automated library management system developed to simplify library operations.
- The system efficiently manages book records, member details, and library resources.
- The system supports member registration and maintenance of member information.
- The project is developed using the C programming language with a menu-driven approach.
- File handling is used to store and retrieve book and member data efficiently.
- AutoLibris reduces manual record-keeping, minimizes human errors, and saves administrative time.

OBJECTIVE:

- To develop an automated library management system for efficient management of books and member records.
- To automate core library operations such as adding, searching, updating, issuing, and returning books.
- To maintain accurate and secure digital records using file handling techniques in C.
  To reduce manual effort and minimize errors in library management through automation.
- To provide a simple, user-friendly interface for librarians to manage library resources efficiently.
- To design a scalable and reliable system that can be enhanced with databases, GUI, cloud services, and AI features in the future.

Literature Survey:

S.No.	Author(s) & Year	Title	Methodology/Technology	Key Findings	Research Gap
1	Sharma, S. & Verma, B. K. (2025)	Intelligent Library Management System	Intelligent automation, digital cataloging	Improved library efficiency through automation and reduced manual work.	Limited AI-based recommendation and predictive analytics.
2	Bhati, P., Komal, Prince & Saxena, S. (2024)	Modern Library Management System: A Web-Based Framework for Resource Automation and User Oversight	Web application, database management	Enabled online access, resource tracking, and user management.	Focused mainly on web implementation with limited offline support.
3	More, N. S. (2024)	Automation and Intelligence in Libraries: A Holistic Review	Review of intelligent library technologies	Highlighted the importance of automation, AI, and digital transformation in libraries.	Did not provide a practical implementation framework.
4	Gangula, R., Bhakar, R. & Amarachinta, P. (2023)	Integration of Library Automation to a Mobile Application with User Friendly Interface	Mobile application, digital library	Improved accessibility through mobile-based library services.	Limited administrative and analytics features.
5	Asid, B. A. (2025)	Library Automation System of Academic Libraries: A Multicultural Paradigm	Academic library automation	Demonstrated automation improves efficiency and user satisfaction.	Lacked intelligent search and advanced reporting capabilities.
6	Dande, P., Newaskar, A., Mandale, D., Thakar, A. & Patil, R. A. (2023)	Smart Library Management System Utilizing RFID Technology	RFID technology	Automated book issue/return process and inventory tracking.	Requires additional hardware, increasing implementation cost.

Summary Of Literature Survey:

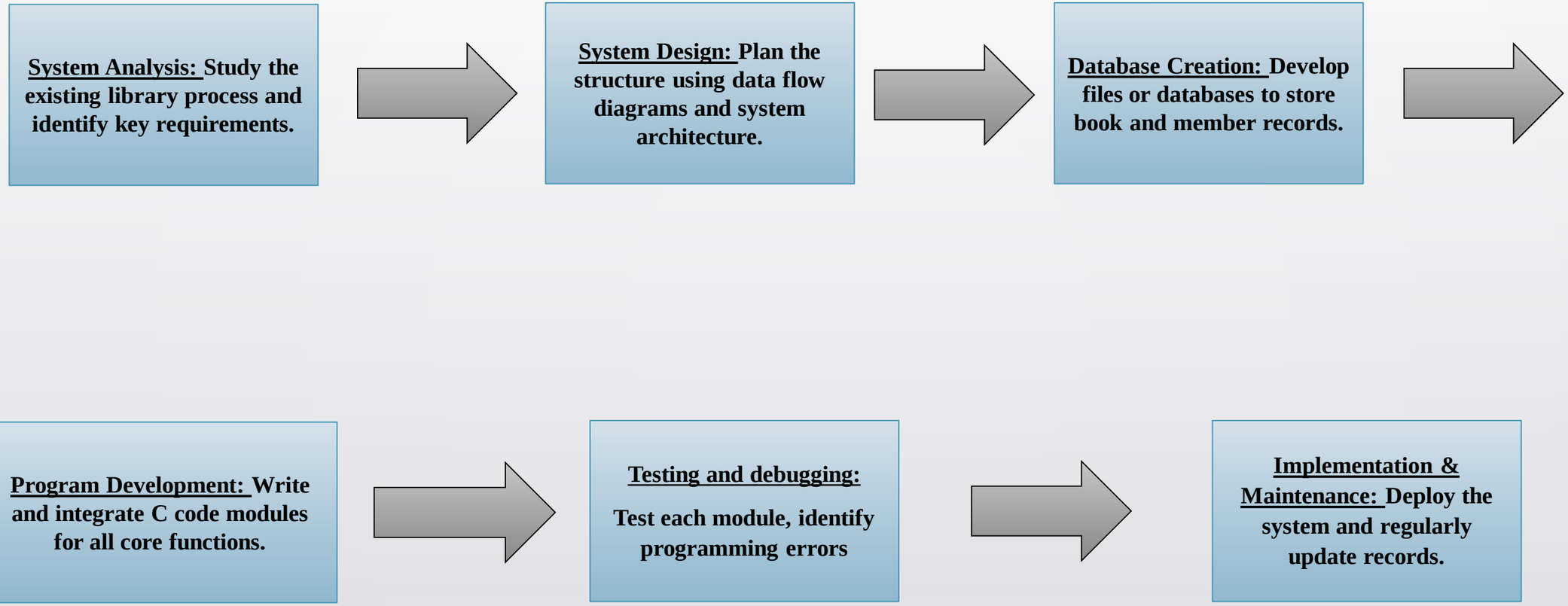
- Traditional library management relies heavily on manual record-keeping, making it time-consuming and prone to human errors.
- Managing books and member information becomes difficult as the library collection grows, leading to inefficient data organization and retrieval.
- Tracking book availability, issuing, and returning processes manually often results in delays, duplicate records, and inaccurate inventory.
- The absence of a centralized digital management system reduces operational efficiency and makes maintaining library records challenging.

Proposed System & Novelty:

- Automates major library management activities using C programming.
- Uses **structures** to organize user and book records.
- Uses **file handling** for permanent storage and retrieval of records.
- Provides modules for **user management, book management, issue/return, file management, and report generation**.
- Reduces manual record-keeping and improves data organization.
- Enables faster searching, updating, issuing, and returning of books.
- Helps maintain accurate book availability and transaction records.

- AutoLibris successfully automates essential library management operations.
- Demonstrates practical use of **C programming, structures, pointers, and file handling**.
- Provides efficient management of books, users, and transactions.
- Reduces manual work and improves record accuracy.
- The modular architecture makes future improvements easier.
- Can be enhanced with **databases, GUI/web interfaces, cloud services, AI recommendations, and advanced security**.

Block Diagram:



REFERENCES

1. Sharma, S., & Verma, B. K. (2025). *Intelligent Library Management System.*
2. Bhati, P., Komal, Prince, & Saxena, S. (2024). *Modern Library Management System: A Web-Based Framework for Resource Automation and User Oversight.*
3. More, N. S. (2024). *Automation and Intelligence in Libraries: A Holistic Review.*
4. Gangula, R., Bhakar, R., & Amarachinta, P. (2021). *Integration of Library Automation to a Mobile Application with User Friendly Interface.*
5. Asid, B. A. (2020). *Library Automation System of Academic Libraries: A Multicultural Paradigm.*


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